

A Reflexive-Subspace Counterexample to Question 5.1 of arXiv:1606.05999

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22 July 2026

Status. Candidate full negative answer, likely valid, pending human review. The example does not merely obstruct distortion $1 + \varepsilon$: the resulting little-Lipschitz space admits no isomorphic embedding into c_0 at all.

Source Question

García-Lirola, Petitjean, and Rueda Zoca ask the following in Question 5.1 of [1], on source PDF page 20:

Let M and τ be under the hypothesis of Theorem 2.9. Is $\text{lip}_\tau(M)$ $(1 + \varepsilon)$ -isometric to a subspace of c_0 for every $\varepsilon > 0$?

provides the norm-denseness of the set of those linear forms which attain their norm. But according to Lemma 4.3, if f attains its norm as linear form on $\mathcal{F}(M) \hat{\otimes}_\pi X^*$, then f also attains its norm as an operator defined on $\mathcal{F}(M)$. This provides the result. Finally, last assertion follows from Proposition 4.2. ■

5. SOME REMARKS AND OPEN QUESTIONS

Note that in [6] it is proved that, given a proper metric space M , then $S(M)$ is $(1 + \varepsilon)$ -isometric to a subspace of c_0 for every $\varepsilon > 0$. So, we wonder.

Question 5.1. Let M and τ be under the hypothesis of Theorem 2.9. Is $\text{lip}_\tau(M)$ $(1 + \varepsilon)$ -isometric to a subspace of c_0 for every $\varepsilon > 0$?

Note that an affirmative answer to above question would imply that $\text{lip}_\tau(M)$ would be an M -embedded space. This would have two implications. On the one hand, given a Banach space X such that $\mathcal{F}(M)$ or X^* has (AP), $\text{lip}_\tau(M, X) = \text{lip}_\tau(M) \hat{\otimes}_\varepsilon X$ would be an unconditionally almost square Banach space [11] and, consequently, it would not be a dual Banach space, extending Proposition 2.11. On the other hand, such a metric space M would satisfy the thesis of Proposition 4.1.

Another question from Section 3 is the following.

Source PDF page 20, including the complete statement of Question 5.1.

The paper works over the real field. Its standing setup for Theorem 2.9 is a bounded pointed metric space (M, d) with a compact topology τ for which d is τ -lower semicontinuous. The theorem additionally assumes that M is separable and complete, that τ is metrizable, and that

$$\forall u, v \in M \ \forall \eta > 0 \ \exists f \in B_{\text{lip}_\tau(M)} : |f(u) - f(v)| \geq d(u, v) - \eta. \quad (\text{P})$$

Here $\text{lip}_\tau(M)$ consists of the pointed τ -continuous Lipschitz functions satisfying

$$\lim_{r \downarrow 0} \sup_{0 < d(u,v) < r} \frac{|f(u) - f(v)|}{d(u,v)} = 0.$$

Proof Intuition

Snowflaking a norm by the square root turns every bounded linear functional into a little-Lipschitz function: a norm-linear increment is quadratic in the new distance. On the weak unit ball of ℓ_2 , all these linear functionals are also weakly continuous. Consequently $\text{lip}_\tau(M)$ contains a closed copy of the whole reflexive space ℓ_2 .

This is incompatible with an embedding into c_0 . One standard way to see the obstruction is that c_0 has the hereditary Dunford–Pettis property. If a closed copy of ℓ_2 lived in c_0 , that copy would have the Dunford–Pettis property; but its identity is weakly compact by reflexivity and is not completely continuous. Thus the source question fails in a much stronger form than requested.

The Counterexample

Theorem 1. *There is a pair (M, τ) satisfying all hypotheses of Theorem 2.9 in [1] such that $\text{lip}_\tau(M)$ is not isomorphic to a subspace of c_0 . In particular, Question 5.1 has a negative answer.*

Proof. Let $H = \ell_2$ over $\mathbb{R}$, put $M = B_H$, and use 0 as the base point. Let τ be the weak topology on B_H and define

$$d(u, v) = \|u - v\|_2^{1/2} \quad (u, v \in B_H).$$

We first audit the hypotheses. The function $\omega(t) = \sqrt{t}$ is a nontrivial gauge in the terminology of [1]: it is continuous, increasing, and subadditive, it vanishes at zero, it satisfies $\omega(t) \geq t$ for $0 \leq t \leq 1$, and $\omega(t)/t \rightarrow \infty$ as $t \downarrow 0$. The sentence introducing the paper’s corollary immediately after Theorem 2.9 explicitly identifies the following as a special case:

$$(B_{X^*}, \omega \circ \|\cdot\|)$$

with its weak-star topology. Taking $X = H$ and identifying $X^* = H$ places our example in the family highlighted by the authors. We nevertheless verify property (P) directly below, so the counterexample does not depend on that identification.

For completeness, the remaining topological and metric hypotheses are also immediate. The ball B_H is weakly compact because H is reflexive, and its weak topology is metrizable because $H^* = H$ is separable. The snowflaked metric is bounded and separable. It is complete because a d -Cauchy sequence is norm Cauchy. Finally, the norm is weakly lower semicontinuous, so $(u, v) \mapsto d(u, v)$ is lower semicontinuous for the product weak topology.

We now prove property (P). Fix $u, v \in B_H$ with $u \neq v$, put $\delta = \|u - v\|_2$, and choose $h \in S_H$ so that $\langle u - v, h \rangle = \delta$. For $\rho > 0$ define

$$\phi_\rho(s) = \sqrt{s_+ + \rho} - \sqrt{\rho}, \quad s_+ = \max\{s, 0\}.$$

The function ϕ_ρ is globally Lipschitz in the ordinary metric, with constant at most $1/(2\sqrt{\rho})$. It also satisfies

$$|\phi_\rho(s) - \phi_\rho(t)| \leq \sqrt{|s - t|} \quad (s, t \in \mathbb{R}). \quad (1)$$

Indeed, this is the usual square-root estimate if $s, t \geq 0$, it is trivial if $s, t \leq 0$, and if $s \leq 0 \leq t$, then

$$\phi_\rho(t) = \sqrt{t + \rho} - \sqrt{\rho} \leq \sqrt{t} \leq \sqrt{|t - s|}.$$

Set

$$f_\rho(z) = \phi_\rho(\langle z - v, h \rangle) - \phi_\rho(\langle -v, h \rangle), \quad z \in B_H.$$

Then $f_\rho(0) = 0$, and f_ρ is weakly continuous. From (1),

$$|f_\rho(z) - f_\rho(w)| \leq \sqrt{|\langle z - w, h \rangle|} \leq \sqrt{\|z - w\|_2} = d(z, w),$$

so $\|f_\rho\|_{Lip} \leq 1$. Its ordinary Lipschitz bound gives

$$\frac{|f_\rho(z) - f_\rho(w)|}{d(z, w)} \leq \frac{1}{2\sqrt{\rho}} d(z, w),$$

which tends uniformly to zero with $d(z, w)$. Hence $f_\rho \in B_{lip_\tau(M)}$. Finally,

$$f_\rho(u) - f_\rho(v) = \sqrt{\delta + \rho} - \sqrt{\rho} \longrightarrow \sqrt{\delta} = d(u, v) \quad (\rho \downarrow 0).$$

This proves property (P); the case $u = v$ is immediate.

For $x \in H$, define $Jx : M \rightarrow \mathbb{R}$ by

$$(Jx)(u) = \langle u, x \rangle.$$

This function vanishes at the base point and is weakly continuous. Moreover, for $u \neq v$,

$$\begin{aligned} \frac{|(Jx)(u) - (Jx)(v)|}{d(u, v)} &\leq \frac{\|x\|_2 \|u - v\|_2}{\|u - v\|_2^{1/2}} \\ &= \|x\|_2 \|u - v\|_2^{1/2} = \|x\|_2 d(u, v). \end{aligned} \tag{2}$$

Because $d(u, v) \leq \sqrt{2}$ on B_H , equation (2) proves that Jx is Lipschitz and

$$\|Jx\|_{lip_\tau(M)} \leq \sqrt{2} \|x\|_2.$$

It also proves the little-Lipschitz condition: if $0 < d(u, v) < r$, then the quotient in (2) is at most $r \|x\|_2$, uniformly in u, v . Thus $Jx \in lip_\tau(M)$.

The reverse norm estimate is just as direct. For $x \neq 0$, take $u = x/\|x\|_2$ and $v = 0$. Then $u \in B_H$, $d(u, 0) = 1$, and

$$\frac{|(Jx)(u) - (Jx)(0)|}{d(u, 0)} = \|x\|_2.$$

Consequently

$$\|x\|_2 \leq \|Jx\|_{lip_\tau(M)} \leq \sqrt{2} \|x\|_2 \quad (x \in H). \tag{3}$$

The linear map $J : H \rightarrow lip_\tau(M)$ is therefore bounded below, so its range is a closed subspace isomorphic to ℓ_2 .

Suppose toward a contradiction that $lip_\tau(M)$ admitted an isomorphic embedding T into c_0 . Then $T(JH)$ would be a closed subspace of c_0 isomorphic to ℓ_2 . But c_0 has the hereditary Dunford–Pettis property [2]; hence $T(JH)$ has the Dunford–Pettis property. Since it is reflexive, its identity operator is weakly compact and therefore, by the Dunford–Pettis property, completely continuous. This is impossible: the image under an isomorphism of the orthonormal basis of ℓ_2 is weakly null while its norms stay bounded away from zero. This contradiction proves that no such T exists. $\square$

Verification Notes

- **Property (P) is explicit.** The functions f_ρ have Lipschitz norm at most one, are weakly continuous and little Lipschitz, and their increments converge to the required snowflake distance. The source’s own weak-star-ball corollary independently confirms that this is the intended class of examples.
- **The snowflake is essential.** In the new metric, $\|u - v\|_2 = d(u, v)^2$, so the Lipschitz quotient of a linear functional is bounded by $\|x\|_2 d(u, v)$ and tends uniformly to zero at small scales.
- **The copy is closed.** The lower inequality in (3), not merely injectivity, guarantees that JH is a closed subspace and permits restriction of any hypothetical embedding into c_0 .
- **The c_0 contradiction is isomorphic.** Hereditary Dunford–Pettis is inherited by every closed subspace of c_0 ; reflexivity and the Dunford–Pettis property are invariant under isomorphism. Thus no choice of distortion can evade the obstruction.
- **No computational dependency.** Every estimate is exact. The argument uses no numerical experiment, asymptotic approximation, or unverified interchange of limits.

Novelty Check and Limitations

A bounded search through 22 July 2026 covered the run registry, solution, attempt, and proof-gap indexes; the local source corpus; the arXiv id, title, and authors; exact wording from Question 5.1; and searches combining `lip_tau(M)`, c_0 , and the paper’s weak-star snowflake notation. No later answer to Question 5.1 was found. This supports, but does not prove, novelty.

The result addresses Question 5.1 only. It makes no claim about Question 5.2 in the same paper, concerning the Schur property of a vector-valued Lipschitz-free space. The recommended human check is the smoothed-ramp proof of property (P), especially (1); the rest of the proof is elementary Banach-space geometry.

References

- [1] L. García-Lirola, C. Petitjean, and A. Rueda Zoca, *On the structure of spaces of vector-valued Lipschitz functions*, Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. **112** (2018), 727–749; arXiv:1606.05999.
- [2] P. Cembranos, *The hereditary Dunford–Pettis property on $C(K, E)$* , Illinois J. Math. **31** (1987), no. 3, 365–373.